

Alexey Semenov

Senior Software Engineer

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PROFILE

Senior Software Engineer with 12+ years building scalable web applications, microservices, and cloud infrastructure across IoT, fintech, and government domains. Currently leading engineering at an early-stage solar mobility company, owning everything from customer-facing products to infrastructure-as-code and AI-assisted development workflows. Proven track record of taking products from hardware prototype to production in under a year, designing OAuth 2.0 authentication systems, and mentoring engineers across teams. Polyglot across TypeScript, Go, Python, and Kotlin, with deep experience on AWS and GCP.

EXPERIENCE

Sono Solar

Lead Software Developer

Munich, Germany
May 2025 – Present

- Lead engineering for a small, early-stage team building the software platform that manages fleets of solar installations across multiple vehicle types — owning customer-facing products, internal tooling, and connected-vehicle infrastructure end to end.
- **Built and shipped** two production React web apps from the ground up: an installation tool used by field engineers, and a dashboard where vehicle owners and fleet operators track solar generation and usage in real time.
- **Provisioned** a complete, production-ready white-label clone of the entire platform infrastructure using Terraform modules, provisioned automatically and running live as a sales enablement asset for prospective enterprise customers.
- **Integrated** AI tooling throughout the development pipeline — automated code review and PR generation, test generation and coverage, and feature scaffolding — measurably accelerating delivery for a lean team.
- Migrated the company's existing infrastructure to infrastructure-as-code on GCP, and reduced cloud spend by auditing and decommissioning unused services.
- Introduced monitoring and user-tracking tooling to surface customer behavior and close the product feedback loop; ran user tests on new features.
- Mentor and guide a small internal tech team (1–3 engineers) and advise external partners on technical solutions.

tado° GmbH

Senior Software Engineer

Munich, Germany
May 2018 – May 2025

- **Delivered** a brand-new product generation from scratch — from bare hardware prototypes to an install-ready product with core features in under a year, built on top of unfinished Matter protocol specifications. High-pressure, high-ownership delivery that the team estimated would otherwise have slipped by a year or more.
- Redesigned the installation flow for the original tado° product generation from the ground up, improving onboarding and setup performance by 30%.
- Designed, implemented, and operated an OAuth 2.0 authentication server.
- Built and maintained integrations with voice assistants and automation platforms, including Alexa, Google Home, and IFTTT.
- Designed and developed the tado° Professional app in Flutter for installers.
- Led infrastructure development with a focus on CI pipelines and cloud solutions on AWS; managed server infrastructure as part of on-call rotation.
- Served as a senior technical anchor across the org: ran regular workshops and meetings for colleagues, frequently demoed engineering work and data analytics, and taught Python, SQL, and software fundamentals to designers, support, and other non-engineers.

- Set up self-serve analytics environments (GitHub-based) so non-engineers could run their own data queries on demand, and collaborated closely with the data science team on data storage and accessibility; collected and analyzed metrics to guide product decisions.

RBKmoney

Software Developer

Moscow, Russia
Aug 2016 – May 2018

- Developed an online payment system with a focus on reliability and security.
- Built and maintained microservices for improved scalability and performance.
- Contributed to infrastructure automation using Docker and SaltStack.

Altarix

Software Developer

Moscow, Russia
Mar 2016 – Aug 2016

- Developed a government analytics system to improve data access and decision-making, and tuned internal search-engine performance for faster, more accurate results.
- Built microservices and web applications to support system scalability.

Lanit

Team Lead / Software Developer

Moscow, Russia
May 2013 – Mar 2016

- Led development of large-scale, nationwide public-sector information systems, coordinating multiple development teams totaling roughly 20–40 engineers.
- Managed and mentored engineers across teams, owning project timelines and delivery quality.
- Built MVPs, conducted technology analysis to guide direction, and defined APIs for external services.

Beeline (VimpelCom)

Technical Support

Moscow, Russia
Nov 2012 – May 2013

- Monitored and maintained critical services across day and night shifts, ensuring high availability through proactive support, scripting, and issue escalation.

EDUCATION

National Research Nuclear University MEPhI

Master's Degree in Cybernetics — Computer Systems & Information Technologies

Moscow, Russia
Jan 2013

SKILLS

Languages: TypeScript · JavaScript · Go · Python · Kotlin · Java · Scala · Groovy

Cloud / DevOps: GCP · AWS · Terraform · Docker · GitHub Actions

Databases: PostgreSQL · MongoDB · DynamoDB · Elasticsearch

Frontend / Mobile: React · Flutter

Other: OAuth 2.0 · Matter protocol · RabbitMQ · FastAPI · Microservices · AI-assisted dev tooling

CERTIFICATIONS & ADDITIONAL EDUCATION

- AWS Certified Solutions Architect – Professional; Building Data Lakes on AWS
- MongoDB for Java Developers (M101J); Spring & XSL certificates
- Functional Programming in Scala — course by Martin Odersky; Unix courses (Yandex)

PROJECTS

QuantumFate — quantumfate.com

- Designed, built, and actively maintain a live web product end to end.

Screenshot Guessing Game

- Co-created a game where players identify video games from a single screenshot. Stack: React, Python, FastAPI.